

VRIDHI ERP

OPERATIONS & TECHNICAL DOCUMENTATION

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Vridhi ERP — Enterprise Operations Portal

A modern, high-performance monorepo-based ERP + CRM portal designed specifically for trade, sales, inventory replenishment, GST billing, and customer relationships in the Indian market.

Key Modules & Capabilities

1. Customer Relationship (CRM) & Follow-ups

- Customer Directory**: Profiles containing contact details, address states (for GST calculations), and tax registration IDs (GSTIN).
- Follow-up Checklist**: Unified scheduler tracking customer actions segmented by **Overdue**, **Today**, and **Upcoming** (next 7 days).
- Interaction Logs**: Logging panel enabling sales agents to log phone calls, emails, and meetings while scheduling next follow-up alerts in a single step.

2. Products & Inventory Management

- Master Catalog**: SKU records, warehouse location categories, and product unit prices.
- Safety Stock Alerts**: Automatic alert flags for products dropping below safe replenishment thresholds.
- Reorder Suggestions**: Automatically calculates optimal order quantities using the replenishment algorithm:
$$\text{Suggested Reorder Qty} = (\text{Safety Stock} \times 2) - \text{Current On-hand Stock}$$

3. Vendors & Procurement

- Supplier Registry**: Database storing partner contact persons, email addresses, and GSTINs.
- Purchase Orders (PO)**: Multi-line item PO builder with status states (`DRAFT` $\rightarrow$ `SENT` $\rightarrow$ `COMPLETED` / `CANCELLED`).
- Shipment Receipt Logs**: Allows logging of partial vendor deliveries, linking to original PO lines and dynamically incrementing stock balances.

4. Sales Challans & Invoicing

- Delivery Challans**: Log customer orders in `DRAFT` status and transactionally deduct stock when transitioned to `CONFIRMED`.
- Tax Invoice Engine**: Dynamically converts confirmed challans into tax invoices.
- GST Calculation Engine**:
- Intra-State**: Split taxes (CGST + SGST) for transactions inside the home state.
- Inter-State**: Unified tax (IGST) for customer locations in other states.
- Ledger & Payment Tracking**: Supports partial payments, remaining balance checks, and logs payments against invoices.

5. Documents & Notifications

- PDF Document Export**: Exports tax invoices, PO forms, and delivery challans using custom-formatted layouts built with PDFKit.
- SMTP Email Delivery**: Sends PDF attachments in the background asynchronously without blocking user response cycles. Audits logs inside the `EmailLog` database table.

Tech Stack

- **Monorepo Architecture**: npm Workspaces
- **Frontend**: React, Vite, TypeScript, Vanilla CSS, TanStack Query, Zustand, Axios
- **Backend**: Node.js, Express.js, TypeScript, SQLite
- **Database ORM**: Prisma ORM
- **Emailing**: Nodemailer

🚪» Local Setup & Commands

Prerequisites

- Node.js v18+ & npm

Setup Steps

1. **Install dependencies**:

```
```bash
```

```
npm install
```

```
```
```

2. **Initialize Database**:

Sync schema changes and run database seeds:

```
```bash
```

```
npm run db:push --workspace=apps/backend
```

```
npm run db:seed --workspace=apps/backend
```

```
```
```

3. **Run Development Servers**:

Starts backend on port `5000` and frontend on port `5173`:

```
```bash
```

```
npm run dev
```

```
```
```

🚪👤 Seeding User Profiles

| Profile | Role | Email Address | Password |
|----------------------|------|----------------------|-------------|
| **Admin Manager** | | `admin@demo.com` | `Passw0rd!` |
| **Sales Executive** | | `sales@demo.com` | `Passw0rd!` |
| **Warehouse Keeper** | | `warehouse@demo.com` | `Passw0rd!` |
| **Accounts Officer** | | `accounts@demo.com` | `Passw0rd!` |

🚪➡️ Running Test Suites

Run integration tests verifying the transaction engines, GST calculations, and background SMTP handlers:

```
```bash
```

```
npm run test --workspace=apps/backend
```

```
```
```

Current test suite results: **54 tests passed successfully** with zero regressions.